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The low carbon transformation of cities and corporate investment efficiency: Evidence from a quasi-natural experiment

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ABSTRACT

This paper studies the impact of low-carbon city pilot (LCCP) policy on corporate investment efficiency based on data from Chinese A-share companies from 2007 to 2019. Our results indicate that LCCP policy significantly reduces corporate investment efficiency by increasing the environmental uncertainty faced by corporations, our fundamental conclusion remains robust after a series of robustness tests. Through heterogeneous analysis, we find that LCCP policy has a more obvious negative impact on corporations with lower diversification levels, non-state-owned corporations within high-carbon emitting industries and those located in high-carbon emitting cities. Further analysis suggests that local governments can alleviate the policy's negative impact on local corporations by increasing their response to the LCCP policy. Overall, this study provides theoretical evidence for a comprehensive examination of the LCCP policy, contributing to China's early achievement of carbon peaking and carbon neutrality as well as providing theoretical guidance for other countries.

1. Introduction

The increasingly severe global climate crisis is a significant issue that urgently requires the attention of all humankind (Liu et al., 2023; Zhang and Ge, 2024). Low-carbon development is universally recognized as a necessary approach to addressing this crisis (Wang et al., 2022). China has repeatedly stated its commitment to achieving carbon peak before 2030 and determination to achieve carbon neutrality before 2060 on the international stage (Zhang and Wang., 2021; Deng et al., 2023). In China, over 80 % of carbon dioxide is emitted by cities (Yin et al., 2023). Facilitating the low-carbon transformation of cities is a crucial step in achieving China's "3060"¹ goal. The low-carbon city pilot (LCCP) policy serves as a key driving force behind urban low-carbon transformation.

The LCCP policy, introduced by China, aims to fully mobilize the enthusiasm of all stakeholders and to continuously accumulate experiences in the low carbon transformation of various aspects (Wang, Shao, et al., 2023). This policy was implemented in 2010 and its scope was expanded twice in 2012 and 2017. Currently, the LCCP policy has covered over one hundred cities nationwide. The Chinese Ministry of Ecology and Environment stated in the "National Low-Carbon City Pilot Program Progress Assessment Report"² that the LCCP policy has achieved positive results and has accumulated valuable experience for local green and low-carbon development. Existing literature also indicates that the LCCP policy has made significant contributions in various aspects such as controlling

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E-mail addresses: zzxplus@163.com (Z. Zhang), 775195543@xtu.edu.cn (Z. Ge), sxl2194@case.edu (S. Liu).¹ 3060 refers to the goals of achieving carbon peak by 2030 and carbon neutrality by 2060.² https://www.mee.gov.cn/ywgz/ydqhbh/wsqtz/202307/t20230713_1036161.shtml<https://doi.org/10.1016/j.asieco.2024.101843>

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carbon emissions, improving air quality and promoting green technological innovation (Hou et al., 2023; Yang et al., 2023; Yin et al., 2023). Nevertheless, the cost of environmental compliance is high (Palmer et al., 1995). Environmental regulations may have a range of negative impacts on corporations, such as increasing compliance costs (Li, Wu, et al., 2024), elevating the external environmental uncertainty faced by corporations (Rodriguez Lopez et al., 2017) and discouraging investment (Li et al., 2021), which can subsequently disrupt normal production and business activities. A natural question arises: does the LCCP policy bring these negative impacts to corporations?

Studying macro policies at the micro level is more conducive to evaluating the effectiveness of policies (Zhou and Zhao, 2022). Environmental regulation policies can directly impact the investment behavior of businesses (Leiter et al., 2011). Companies may have to increase socially responsible investments due to governmental pressures (Li, Liu, et al., 2022), even if these investments may be inefficient. Additionally, the previous research on environmental regulation and corporate performance mostly reflects the historical performance of companies, while corporate investment efficiency can, to some extent, reflect the future operational capabilities of companies (Dong et al., 2020). To more accurately assess the impact of the LCCP policy on corporate investment, this paper focuses on corporate investment efficiency. It measures the alignment between a corporate investment activities and business development, which is crucial for the corporate survival and growth (Xie et al., 2023).

What kind of impact does the LCCP policy have on corporate investment efficiency? The answer to this question may affect the policy evaluators' assessment of this policy, which has lasted for more than a decade, covering over a hundred cities in China. If the LCCP policy can not only reduce urban carbon dioxide emission levels but also improve corporate investment efficiency, policymakers can confidently promote the low-carbon pilot experience to other cities. However, if the LCCP policy comes at the expense of sacrificing corporate investment efficiency to reduce carbon emissions at the urban level, relevant parties need to focus on further refine this policy and thoroughly assess the potential risks associated with its widespread promotion. The significance of this paper lies in expanding the research on the effects of LCCP policy at the corporate level, while also providing a theoretical basis for policymakers and implementors.

We conduct regression analysis using annual data on Chinese A-share listed companies for the period 2007–2019. The results reveal that LCCP policy reduces corporate investment efficiency. Different companies in the sample have different responses to LCCP policy, depending on diversification, located cities and ownership type. LCCP policy has a more obvious negative effect on the corporate investment efficiency of lower diversification levels, non-state-owned enterprises within high-carbon emitting industries and those located in high-carbon emitting cities. Furthermore, LCCP policy reduces corporate investment efficiency by increasing the environmental uncertainty faced by corporations, which affect normal production and operation of the firms. Finally, further analysis suggests that local governments can alleviate the policy's negative impact on local corporations by increasing their response to the LCCP policy.

This study contributes to the existing literature in several ways. First, this paper contributes to a comprehensive evaluation of the effects of the LCCP policy. Previous literature has often focused on the positive effects of the LCCP policy (Chen et al., 2023; Wen et al., 2023; Zhao et al., 2023). However, this paper is the first to discover that the LCCP policy could have certain negative impacts on corporate investment efficiency. This finding facilitates a more accurate assessment of the policy's effectiveness and provides a new consideration perspective for the broader implementation of this pilot policy. Second, this paper enriches the research on the factors influencing corporate investment efficiency. Existing literature has examined the factors influencing corporate investment efficiency from multiple perspectives (Chen et al., 2022; Du et al., 2023; Xie et al., 2023). However, there is still limited research exploring the factors affecting corporate investment efficiency from the perspective of low-carbon development. This paper examines the impact of the LCCP policy on corporate investment efficiency, providing empirical evidence from China for the costly regulation hypothesis and enriching the relevant theoretical research. Last, this paper contributes to the improvement and optimization of environmental protection policies. There is a gap between the formulation of the LCCP policy and its successful implementation at the city level, and bridging this gap is crucial (Guo, 2023). This paper is the first to use data from the "National Low-Carbon City Pilot Progress Assessment Report" released by the Chinese government to demonstrate local governments can eliminate or even reverse the negative impact of the LCCP policy on local corporate investment efficiency by enhancing their responsiveness to the policy. This provides a new perspective for the improvement and optimization of environmental protection policies.

The remainder of this paper is organized as follows. Section 2 introduces literature review and hypothesis development. Section 3 is the research design. Section 4 is the baseline regression results, mechanism analysis and robustness tests. Section 5 is the heterogeneity analysis. Section 6 is further analysis. Section 7 is the Conclusion and policy recommendation.

2. Literature review and hypothesis development

2.1. Literature review

Scholars' analysis of policy effects can be divided into macro and micro perspectives. The LCCP policy is a comprehensive and long-lasting environmental regulation policy, since its enactment, it has attracted widespread attention and has been extensively studied in a considerable amount of literature. On a macro level, the LCCP policy has effectively reduced carbon dioxide emissions, improved energy efficiency, and lowered carbon emission intensity (Hou et al., 2023; Dong et al., 2023). Additionally, the LCCP policy has positive effect in enhancing urban air quality, reducing PM2.5 concentration, improving the efficiency of synergistic governance efficiency of carbon and smog (Li, Bai, et al., 2022; Chen et al., 2023; Yang et al., 2023). In addition, Wang, Liu et al. (2023) found that the LCCP policy can enhance enterprises' labor demand by encouraging or compelling them to participate in front-end environmental governance. At the micro level, a substantial body of literature, based on the Porter hypothesis, has investigated the LCCP policy. Some

scholars argue that the LCCP policy enhances the green innovation capabilities of businesses (Yin et al., 2023). Wen et al. (2023) suggests that the LCCP policy significantly improves the total factor productivity of enterprises. Zhao et al. (2023) believes that the LCCP policy can promote the digital transformation of businesses, while Liu and Lv (2023) found that the LCCP policy increases the financialization level of enterprises. Additionally, Li, Zheng et al. (2024) argue that the LCCP policy enhances the environmental performance of businesses. All of these studies fully demonstrate the positive role of the LCCP policy in social development and environmental protection. While there is limited research on the negative impacts of the LCCP policy, this does not imply that environmental regulation policies always achieve environmental goals without affecting economic growth in the short term. For example, Xie et al. (2023) find that the implementation of environmental taxes significantly reduces the investment efficiency of enterprises. Unfortunately, there is currently a lack of literature finding the negative impacts of the LCCP policy at the micro level.

As Segura et al. (2018) stated, policymakers need to consider not only the emission reduction goals of low-carbon policies but also their impact on the economic performance of businesses. Corporate investment efficiency is a crucial indicator for business operations and management, it can fairly and accurately reflect a corporate development capability and operational status. Existing literature has examined factors influencing corporate investment efficiency from both internal and external perspectives. Biddle et al. (2009) reported on the relationship between corporate financial reporting quality and corporate investment efficiency. High-quality financial reporting improves the overall investment efficiency of corporations by reducing over-investment in cash-abundant corporations and mitigating under-investment in cash-constrained corporations. Wang and Yan (2023) argued that improving employee treatment can reinforce corporate investment efficiency through improved internal control and alleviating financing constraints. Bai, He et al. (2023) found that institutional common ownership can strengthen corporate investment efficiency by reducing over investment. In the literature on corporate investment efficiency, most studies focus on internal factors, with only a few examining the impact of external factors on corporate investment efficiency from an external perspective. Policy intervention is a representative research perspective (Chen et al., 2011). Wang and Zhu (2023) found that increasing export rebates can reduce the stickiness of corporate taxes and thereby improve corporate investment efficiency through the study of tax incentive policies. Zhou and Zhao (2022) treated the revitalization plan launched by China after the 2008 financial crisis as a selective industrial policy and found that selective industrial policies can lower corporate investment efficiency by promoting corporate over-investment. Up to now, there is no consensus in the academic community regarding the impact of such low-carbon policies like the LCCP policy on corporate investments.

2.2. Hypothesis development

The LCCP policy puts forward comprehensive requirements for the low-carbon development of businesses but does not specifically address the specific pathways for corporate transformation. This expands the operational space for management, potentially leading to an increase in the degree of internal information asymmetry and the agency problem within the company. According to neoclassical theory, corporate investment activities should focus on projects with a positive net present value (NPV) and the corporation reaches the optimal investment state when marginal returns equal marginal costs. The agency theory posits that conflicts of interest often exist between shareholders and management and management may not necessarily consider maximizing per-share stock value as the ultimate goal (John et al., 2008). This may lead management to reject value-creating projects or accept projects that lack investment value due to their own interests, thereby reducing the corporate investment efficiency (Aggarwal and Samwick, 2006). The adverse selection and moral hazard issues resulting from information asymmetry often lead to over-investment or under-investment by corporations (Zhou et al., 2023), thereby deviating the corporations from the optimal investment level and reducing their investment efficiency. In addition, Porter and Linde (1995) proposed the well-known Porter hypothesis, suggesting that appropriate environmental regulations can promote corporate innovation, bring "compensating returns" to corporations and enhance corporations' operational conditions (Han et al., 2024). However, Palmer et al. (1995) put forth a starkly different viewpoint—the costly regulation hypothesis. They argued that corporations instinctively seek opportunities to improve financial performance and environmental regulations hinder corporations' pursuit of profits. Under stricter environmental regulations, corporations have to bear higher costs to comply with government supervision, which crowds out necessary investment expenditures and may consequently lead to a decline in investment efficiency (Rassier and Earnhart, 2015). Due to the "compensating returns" mentioned by the Porter hypothesis typically taking a longer period to materialize, we believe that in the short term, the costly regulation hypothesis will hold a more dominant position.

In conclusion, we propose the following hypothesis:

H1. LCCP policy will significantly reduce corporate investment efficiency.

The core objective of the LCCP policy is to reduce carbon emissions, and low carbon transformation is essential for corporate development. Low-carbon transformation will increase corporate compliance costs in the short term (Li, Wu et al., 2024), the costs associated with the government's formulation, implementation and supervision of environmental regulations, which constitute external costs for corporations, will ultimately be borne by the companies (Zhao et al., 2018). The LCCP policy to some extent disrupted the existing plans of the corporations, corporations may need to adjust existing business plans, increase new regulatory and reporting systems to ensure their low-carbon practices comply with policy requirements. The LCCP policy also disrupts corporate original external environment, increasing the uncertainty faced by corporations. With the increasing external environmental uncertainty, corporations may become more cautious, leading to delayed or reduced necessary investment, potentially reducing corporate investment efficiency (Li et al., 2021; Rumokoy et al., 2023; Hu et al., 2023).

Based on the above analysis, this paper proposes the second hypothesis:

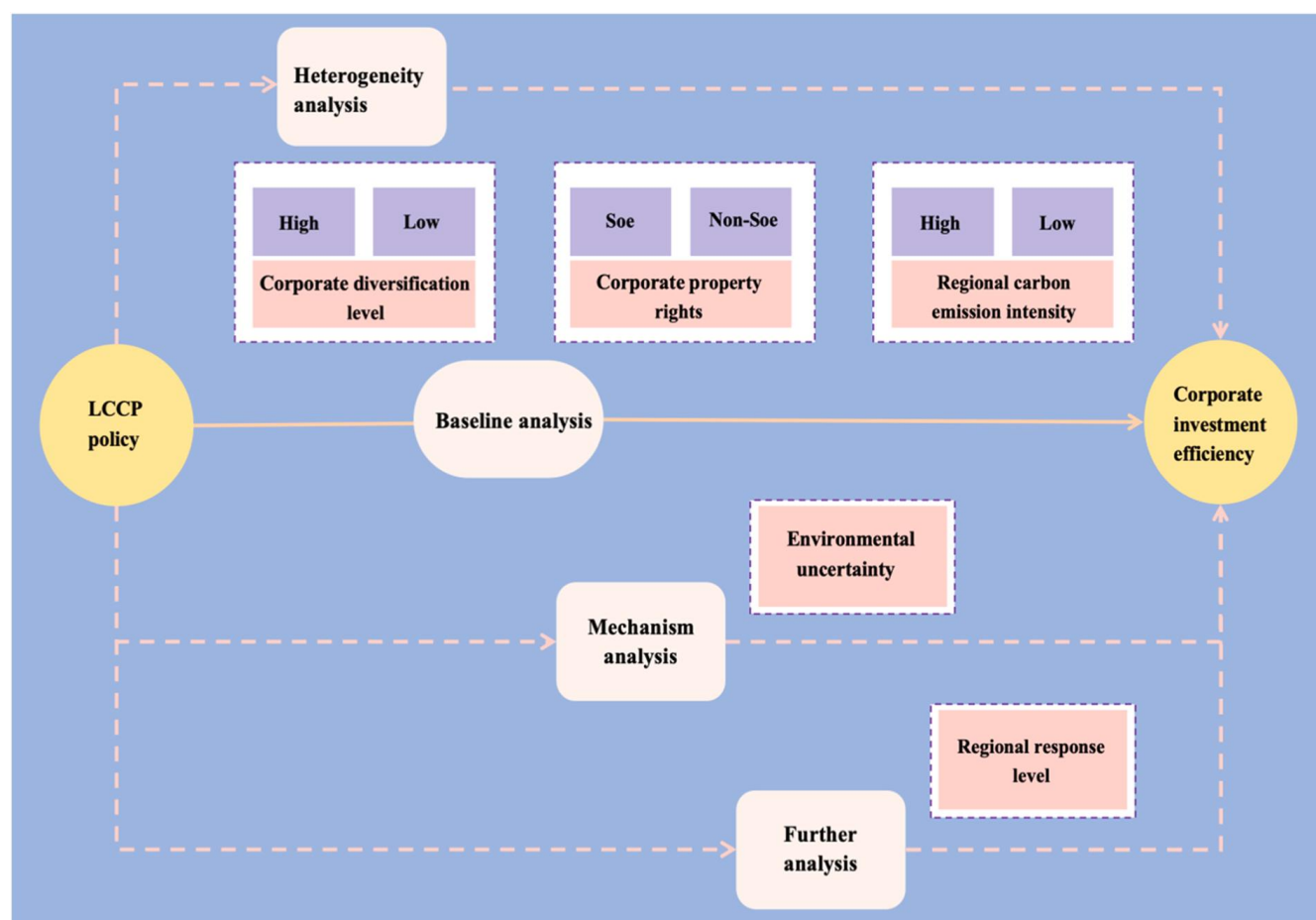


Fig. 1. Research framework.

H2. LCCP policy will affect corporate investment efficiency via its effects on external environment uncertainty.

Under the intervention of the LCCP policy, corporations need to adjust their investment concepts, formulate new investment plans and select new investment targets. In comparison to highly diversified corporations, those with lower diversification have fewer investment options (Wang, 2023). Consequently, they may be more significantly impacted by this policy. The LCCP policy requires pilot cities to effectively control local carbon dioxide emissions, which undoubtedly imposes significant policy shocks on corporations in high-carbon emitting industries. Considering that the LCCP policy eliminates some investment opportunities while creating new ones, the challenges and opportunities faced by corporations of different equity natures are unequal during this process (Yin et al., 2023). Chinese state-owned enterprises are large in scale and have government backing. In contrast, non-state-owned enterprises lack these conditions and may exhibit more negative performance when facing policy shocks. Taking into account the policy information advantages and better policy orientation identification capabilities of state-owned enterprises (SOEs), the impact of the LCCP policy on corporate investment efficiency is likely to vary depending on the different ownership types. LCCP policy requires pilot areas to facilitate urban low-carbon transformation through adjusting industrial structures. In comparison to cities with lower carbon emissions, cities with higher carbon emissions often have more high-carbon emitting industries and corporations, posing greater resistance to urban low-carbon transformation (Dong et al., 2023). Local corporations may need to spend more time, energy and resources to meet the requirements of low-carbon policies, inevitably leading to the suppression of their investment activities, potentially resulting in a negative impact on investment efficiency.

The above discussion leads to our third hypothesis:

H3. The impact of LCCP policy on corporate investment efficiency will be heterogeneous among corporations, depending on diversity level, ownership type and carbon emissions of the city where the corporation is located.

According to the analysis above, we have established the research framework of this paper (Fig. 1).

3. Research design

3.1. Sample and data

In order to investigate the impact of the LCCP policy on corporate investment efficiency, this paper utilizes a sample of Chinese A-

Table 1
Summary statistics.

Variable	N	Mean	SD	Min	p50	Max
InvEff	22357	0.0440	0.0440	0	0.0360	0.815
LCCP	28986	0.484	0.500	0	0	1
Size	28986	22.05	1.318	14.94	21.86	28.64
Roa	28986	0.0370	0.0930	−4.946	0.0390	0.590
Lev	28986	0.428	0.214	0.00800	0.420	4.026
Top1	28986	0.354	0.152	0.00300	0.336	0.900
Board	28986	2.145	0.202	0.693	2.197	2.890
InBoard	28986	0.373	0.0560	0	0.333	0.800
Cash	28986	0.192	0.145	0	0.150	0.993
Fla	28986	0.571	0.211	0.0140	0.587	1
Dual	28986	0.259	0.438	0	0	1
Soe	28986	0.406	0.491	0	0	1

Note: This table reports the summary statistics for variables used in the baseline model.

share listed companies from 2007 to 2019 as the initial research sample. All financial data involved are sourced from the China Stock Market and Accounting Research (CSMAR) database. The regression data were filtered according to the following principles: (1) Using the industry classification announced by the China Securities Regulatory Commission in 2012 as the standard, listed company samples belonging to the financial industry were excluded. (2) Samples of companies marked with ST or *ST during the sample period were excluded. (3) Samples of companies with missing relevant financial data were also excluded. After the aforementioned filtering process, this study finally obtained an unbalanced panel data comprising 3555 listed companies with a total of 28,986 observations.

3.2. Variable selection and description

3.2.1. Corporate investment efficiency

Following the approach of Biddle et al. (2009), we perform an OLS regression on Eq. (1) and use the absolute value of the resulting residuals as the dependent variable (InvEff) in the benchmark regression Eq. (2) of this paper. It should be noted that InvEff is an inverse proxy for corporate investment efficiency. The larger the InvEff, the lower the investment efficiency of the company, and the smaller the InvEff, the higher the investment efficiency (Wang and Zhu, 2023).

$$\text{Invest}_{i,t} = \alpha_1 + \beta_1 \text{Growth}_{i,t-1} + \varepsilon_{i,t} \quad (1)$$

In Eq. (1), Invest represents the investment level of the corporation, we calculated it based on Xie et al. (2023) using a common evaluation method as Invest = (cash spent on purchasing and constructing fixed assets, intangible assets and other long-term assets + net cash paid for acquiring subsidiaries and other operating units - net cash received from disposal of fixed assets, intangible assets and other long-term assets - net cash received from disposal of subsidiaries and other operating units) / total assets. Growth represents the growth opportunities of the corporation, measured in this study by the growth rate of operating profit. α_1 is the intercept term and ε denotes the random disturbance term.

3.2.2. LCCP policy

According to the "Notice on the Pilot Work of Low-Carbon Provinces and Low-Carbon Cities"³ issued by the National Development and Reform Commission, as well as two subsequent documents, it is evident that the three rounds of LCCP policy were respectively launched on July 19, 2010, November 26, 2012 and January 7, 2017. Furthermore, the first round of LCCP policy included 5 provinces and 8 cities; the second round covered 1 province and 28 cities; and the third round involved 41 cities and 4 districts. The specific list of pilot cities can be found in Appendix A. Drawing on the approach of Wang, Liu et al. (2023), this study sets the pilot policies and pilot cities as follows: First, based on the specific implementation dates of the policies, this study sets the start time of the three rounds of LCCP policy as 2011, 2013 and 2017, respectively. Second, if city c participates in the LCCP policy in year t , then city c will be considered as participating in the low-carbon pilot for all years from t onwards. Last, when a province participates in the pilot, all cities within that province are considered as participating in the low-carbon pilot.

Based on the provided information, we have determined the assignment criteria for the core explanatory variable LCCP in this study: if the city c where the corporation i is located is within the scope of the LCCP policy in year t , the dummy variable LCCP is assigned a value of 1; otherwise, LCCP is assigned a value of 0. At the city level, the sample of this study involves a total of 384 cities, including 131 pilot cities and 253 non-pilot cities.

3.2.3. Control variables

This paper draws on previous literature to incorporate a series of control variables into the regression model to enhance the accuracy and reliability of the study. We refer to previous literature (Zhou et al., 2022; Xie et al., 2023; Wang, 2023) and select

³ https://www.ndrc.gov.cn/fggz/tzgg/ggkx/201008/t20100825_1050044.html

Table 2
Correlation analysis.

	InvEff	LCCP	Size	Roa	Lev	Top1	Board	InBoard	Cash	Fla	Dual	Soe
InvEff	1											
LCCP	−0.018***	1										
Size	−0.047***	0.109***	1									
Roa	−0.023***	−0.029***	0.018***	1								
Lev	0.035***	−0.034***	0.466***	−0.336***	1							
Top1	0.00200	−0.038***	0.212***	0.110***	0.049***	1						
Board	−0.034***	−0.114***	0.244***	0.021***	0.159***	0.025***	1					
InBoard	0.019***	0.093***	0.034***	−0.030***	−0.012**	0.044***	−0.490***	1				
Cash	−0.042***	−0.010*	−0.258***	0.180***	−0.399***	0.026***	−0.057***	0.00300	1			
Fla	−0.125***	0.072***	−0.206***	0.099***	−0.110***	−0.00900	−0.143***	0.038***	0.490***	1		
Dual	0.024***	0.084***	−0.165***	0.028***	−0.160***	−0.058***	−0.174***	0.097***	0.116***	0.128***	1	
Soe	−0.036***	−0.106***	0.340***	−0.048***	0.310***	0.214***	0.283***	−0.059***	−0.146***	−0.220***	−0.303***	1

Note: ***, ** and * denote significance at the 1 %, 5 % and 10 % levels, respectively.

Table 3
Baseline results.

VARIABLES	(1)	(2)
	InvEff	InvEff
LCCP	0.002** (1.99)	0.002** (2.01)
Size		−0.004*** (−3.75)
Roa		0.018* (1.68)
Lev		0.019*** (3.98)
Top1		0.022*** (3.73)
Board		−0.000 (−0.01)
InBoard		0.012 (1.23)
Cash		0.013** (2.52)
Fla		−0.041*** (−8.10)
Dual		0.003** (2.39)
Soe		−0.004 (−1.52)
Constant	0.043*** (66.81)	0.139*** (5.43)
Firm FE	YES	YES
Year FE	YES	YES
Observations	22,218	22,218
R-squared	0.281	0.288

Note: T values are in parentheses. ***, ** and * denote significance at the 1 %, 5 % and 10 % levels, respectively.

corporation size (Size), leverage ratio (Lev), return on assets (Roa), current asset ratio (Fla), the shareholding ratio of the largest shareholder (Top1), cash holding ratio (Cash), board size (Board), proportion of independent directors (InBoard), whether the chairman and the general manager are the same person (Dual) and ownership nature (Soe) as firm-level control variables. A detailed description of the control variables can be found in Appendix B.

3.3. Model specification

3.3.1. LCCP policy and corporate investment efficiency

This study constructs the Staggered DID regression Eq. (2) to investigate the relationship between LCCP policy and corporate investment efficiency.

$$\text{InvEff}_{i,t} = \alpha_2 + \beta_2 \text{LCCP}_{i,t} + \gamma_2 \text{Conall}_{i,t} + \text{Firm} + \text{Year} + \varepsilon_{i,t} \quad (2)$$

In Eq. (2), InvEff is the absolute value of the residuals from the regression in Eq. (1) and it serves as an inverse proxy for corporate investment efficiency. LCCP reflects whether corporation's location participated in the LCCP policy. Conall_{i,t} represents a series of control variables at the firm level. Firm and Year represent firm fixed effects and year fixed effects, respectively. α is the intercept term and ε denotes the random disturbance term.

3.4. Descriptive statistics and correlation analysis

The descriptive statistics and correlation analysis of the variables at the corporation level are shown in Table 1 and Table 2, respectively. In the descriptive statistics table, the median of the corporate investment efficiency indicator (InvEff) is less than the mean, indicating that the investment efficiency of most corporations is higher than the average level. The standard deviation of InvEff is equal to the mean and greater than the median, suggesting significant individual differences in corporate investment efficiency. The average value of the dummy variable LCCP is 0.484, indicating that 48.4 % of the sample participated in the LCCP policy, this is comparable to those reported in prior literature (Liu and Lv, 2023; Yin et al., 2023). Regarding the control variables, the average corporation size of the sample companies is 22.05, the leverage ratio is 0.428, the return on assets (ROA) is 0.037 and the average ownership nature is 0.406, indicating that 40.6 % of the sampled corporations are state-owned. Overall, the descriptive statistics of the control variables are generally consistent with those reported in the existing literature (Yin et al., 2023; Zhang, 2023).

In the correlation analysis table, the absolute values of the correlation coefficients between variables are all less than 0.5 and the

Table 4
Mechanism analysis.

VARIABLES	(1)	(2)
	EU	InvEff
LCCP	0.130** (2.46)	
EU		0.009*** (3.35)
Size	−1.014*** (−10.98)	0.005** (2.54)
Roa	−0.130 (−0.46)	0.025** (2.42)
Lev	0.094 (0.38)	0.028*** (8.45)
Top1	0.656** (1.98)	−0.002 (−1.04)
Board	−0.032 (−0.18)	−0.005** (−2.44)
InBoard	0.738** (1.99)	−0.015* (−1.72)
Cash	0.983*** (3.88)	0.015*** (4.08)
Fla	−0.522* (−1.86)	−0.015*** (−4.60)
Dual	0.135*** (2.59)	0.001 (0.74)
Soe	−0.203 (−1.34)	−0.003*** (−3.68)
Constant	23.428*** (11.72)	−0.072 (−1.37)
Firm FE	YES	YES
Year FE	YES	YES
Observations	16,208	16,208
R-squared	0.157	.

Note: T values are in parentheses. ***, ** and * denote significance at the 1 %, 5 % and 10 % levels, respectively.

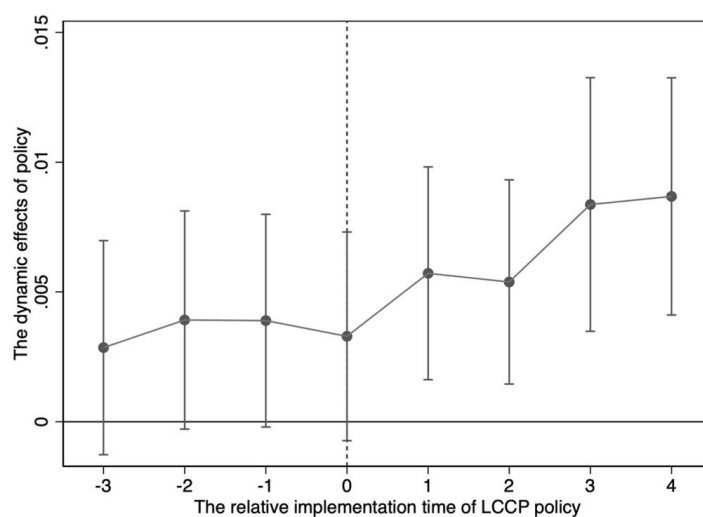


Fig. 2. Parallel trend test.

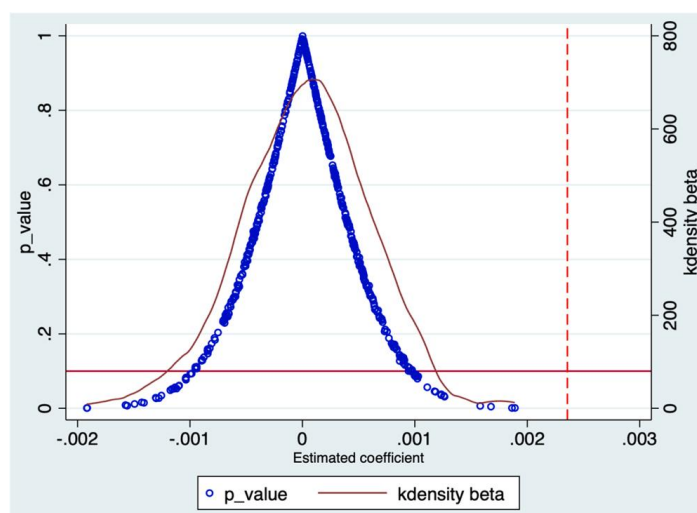


Fig. 3. Placebo test.

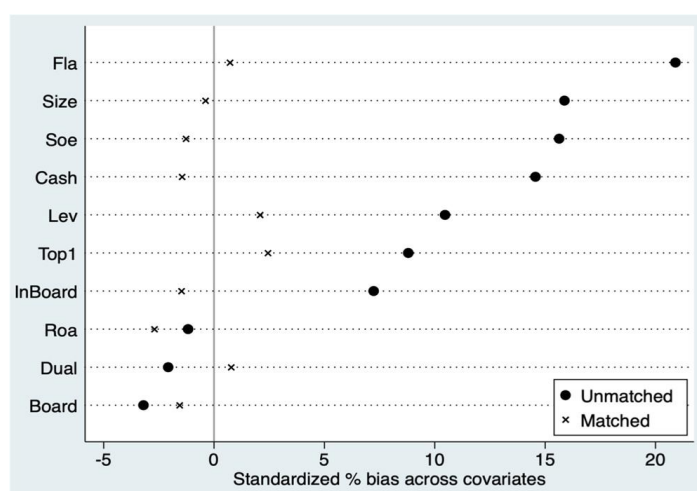


Fig. 4. Balance test.

variance inflation factor (VIF) are all less than 2. Thus, it can be considered that there is no issue of multicollinearity among the variables and the selection of variables is reasonably sound.

4. Empirical results and Robustness tests

4.1. LCCP policy and corporate investment efficiency

The main regression results of this study are shown in Table 3. The first and second columns display the regression results without the control of variables and the results after controlling for variables, respectively. Whether or not control variables are included, the coefficient of LCCP is consistently significant and positive, passing the significance test at the 5 % level. This indicates that compared to non-pilot cities, the LCCP policy significantly increases the level of inefficient investment for corporations within pilot cities. In other words, the LCCP policy significantly reduces the investment efficiency of corporations within pilot cities. This finding validates hypothesis H1 of this study. Our findings regarding the control variables are largely consistent with previous research. For example, Lev is negatively correlated with InvEff (Xie et al., 2023), and Top1 is negatively correlated with InvEff, which is consistent with Yang and Li (2024).

4.2. Mechanism analysis

According to theoretical analysis, we posit that LCCP will reduce corporate investment efficiency by impacting the environmental uncertainty faced by corporations. To examine this mechanism, we employ a mechanism analysis model composed of two parts, following the approach outlined by Di Giuli and Laux (2022). This analytical model conducts causal mediation analysis within an

Table 5
Robustness check I.

	(1) PSM-DID	(2) Changing dependent variable	(3) Excluding ICP policy
VARIABLES	InvEff	InvEff2	InvEff
LCCP	0.002** (2.04)	0.002* (1.68)	0.002* (1.83)
ICP			0.002 (1.15)
Size	−0.004*** (−3.68)	−0.005*** (−7.01)	−0.004*** (−3.77)
Roa	0.029*** (3.61)	0.019*** (5.03)	0.018* (1.69)
Lev	0.017*** (3.67)	0.015*** (4.88)	0.019*** (3.97)
Top1	0.021*** (3.61)	0.019*** (4.04)	0.022*** (3.70)
Board	0.000 (0.05)	0.001 (0.19)	−0.000 (−0.01)
InBoard	0.013 (1.25)	0.009 (1.02)	0.012 (1.24)
Cash	0.011** (2.18)	0.012*** (3.15)	0.013*** (2.50)
Fla	−0.041*** (−8.03)	−0.035*** (−9.98)	−0.041*** (−8.11)
Dual	0.003** (2.33)	0.003*** (2.84)	0.003** (2.41)
Soe	−0.004 (−1.48)	−0.008*** (−3.29)	−0.004 (−1.50)
Constant	0.137*** (5.37)	0.161*** (8.94)	0.139*** (5.42)
Firm FE	YES	YES	YES
Year FE	YES	YES	YES
Observations	22,214	24,546	22,218
R-squared	0.288	0.269	0.288

Note: T values are in parentheses. ***, ** and * denote significance at the 1 %, 5 % and 10 % levels, respectively.

instrumental variable regression framework, which is more robust and reliable compared with traditional mediation effect models (Ho et al., 2023). In this paper, the first part of the mechanism analysis model examines the impact of the LCCP policy on environmental uncertainty faced by corporations. The second part investigates the influence of environmental uncertainty faced by corporations on corporate investment efficiency. The specific mechanism analysis model is as Eq. (3) and Eq. (4):

$$M_{i,t} = \alpha_3 + \beta_3 LCCP_{i,t} + \gamma_3 Conall_{i,t} + Firm + Year + \varepsilon_{i,t} \quad (3)$$

$$InvEff_{i,t} = \alpha_4 + \beta_4 M_{i,t} + \gamma_4 Conall_{i,t} + Firm + Year + \varepsilon_{i,t} \quad (4)$$

In Eq. (3) and Eq. (4), M represents the mechanism variable, with meanings of the remaining variables consistent with Eq. (2). The mechanism variable in this study is the environmental uncertainty faced by firms (EU). We refer to Ghosh and Olsen (2009) by using the coefficient of variation of sales to measure EU. The larger the value of EU, the greater the environmental uncertainty the firm faces. The logic behind Ghosh and Olsen's (2009) definition of EU is that the coefficient of variation of sales measures the ratio of the standard deviation to the mean of a firm's sales. The larger this ratio, the greater the fluctuation in the firm's sales, indicating a more unstable market environment and market volatility directly reflects environmental uncertainty. Furthermore, there is a significant positive correlation between the coefficient of variation of sales and managers' subjective perception of environmental uncertainty. This means that the coefficient of variation of sales is not only theoretically justified but also validated in practice.

The mechanism analysis is shown in Table 4, Results in column (1) demonstrate that LCCP significantly increases environmental uncertainty for corporations at a 5 % significance level. Column (2) indicates, at a 1 % significance level, that as environmental uncertainty faced by corporations rises, the investment efficiency of corporations decreases, signifying a clear negative correlation between environmental uncertainty and corporate investment efficiency. The above results strongly demonstrate the existence of the influence channel of environmental uncertainty. The implementation of the LCCP policy has increased the environmental uncertainty faced by firms, thereby reducing corporate investment efficiency. This finding validates hypothesis H2 of this study.

Table 6
Robustness check II.

	(1)	(2)	(3)
	Excluding CET policy	Shrink 1 %	Shrink 5 %
VARIABLES	InvEff	InvEff	InvEff
LCCP	0.003** (2.15)	0.002* (1.87)	0.001* (1.72)
CET	−0.001 (−0.80)		
Size	−0.004*** (−3.73)	−0.003*** (−3.77)	−0.003*** (−3.92)
Roa	0.018* (1.68)	0.057*** (7.68)	0.070*** (8.50)
Lev	0.019*** (3.99)	0.016*** (4.28)	0.011*** (3.90)
Top1	0.022*** (3.73)	0.016*** (3.22)	0.012*** (2.85)
Board	0.000 (0.01)	−0.000 (−0.14)	−0.002 (−0.72)
InBoard	0.012 (1.23)	0.011 (1.16)	0.004 (0.40)
Cash	0.013** (2.51)	0.006 (1.48)	0.005 (1.50)
Fla	−0.041*** (−8.08)	−0.037*** (−9.82)	−0.029*** (−9.92)
Dual	0.003** (2.40)	0.002** (2.45)	0.002* (1.88)
Soe	−0.004 (−1.52)	−0.003 (−1.47)	−0.002 (−1.23)
Constant	0.139*** (5.41)	0.119*** (5.85)	0.105*** (6.55)
Firm FE	YES	YES	YES
Year FE	YES	YES	YES
Observations	22,218	22,218	22,218
R-squared	0.288	0.302	0.303

Note: T values are in parentheses. ***, ** and * denote significance at the 1 %, 5 % and 10 % levels, respectively.

4.3. Robustness tests

4.3.1. Parallel trend test

Satisfying the parallel trend assumption is a prerequisite for using the difference in difference model. We need to verify that before the implementation of the LCCP policy, there exists a clear parallel trend in the corporate investment efficiency between the control group (LCCP=0) and the experimental group (LCCP=1). Referring to the study by Beck et al. (2010), we have constructed Eq. (5).

$$\text{InvEff}_{i,t} = \alpha_5 + \sum_{-3}^4 \beta_5 \text{LCCP}_{i,t} + \gamma_5 \text{Conall}_{i,t} + \text{Firm} + \text{Year} + \varepsilon_{i,t} \quad (5)$$

This study adopts an event study approach to test the parallel trend assumption between the control group and the experimental group. Specifically, we take the year of the LCCP policy implementation as the base year and examine the parallel trends in the base year, three years prior and four years after. The variables in Eq. (5) have the same meanings as those in Eq. (2). As shown in Fig. 2, before the implementation of the LCCP policy, there is no significant difference in the corporate investment efficiency between the pilot cities and non-pilot cities. However, significant differences emerge between the two groups after the policy implementation. Therefore, we can conclude that the sample used in this study has passed the parallel trend test.

4.3.2. Placebo test

To examine whether the impact of the LCCP policy on corporate investment efficiency is merely coincidental, this study utilizes a placebo test to identify the randomness of the LCCP policy. Referring to the approaches of Chetty et al. (2009), we randomly draw 500 samples based on the distribution of the LCCP policy in the baseline regression to construct "pseudo-policy dummy variables" and re-estimate the regression using Eq. (2). The coefficient and p-value distribution results are shown in Fig. 3. The mean of the regression coefficient of corporate investment efficiency on the "pseudo-policy dummy variables" is close to 0 and approximately follows a normal distribution. The baseline regression coefficient is 0.0024, located in the high tail of the distribution of the placebo test regression coefficients and the majority of p-values are not significant at the 10 % level. These results indicate that the impact of the LCCP policy on corporate investment efficiency is not caused by other random factors, thus confirming the reliability of the results from the baseline regression.

Table 7
Heterogeneity analysis I.

VARIABLES	(1)	(2)
	High diversification	Low diversification
	InvEff	InvEff
LCCP	−0.002 (−0.55)	0.003** (2.23)
Size	−0.003 (−1.00)	−0.003*** (−2.66)
Roa	0.055* (1.93)	0.014 (1.09)
Lev	0.023* (1.90)	0.017*** (3.09)
Top1	0.027* (1.76)	0.020*** (3.13)
Board	0.012 (1.62)	−0.003 (−0.86)
InBoard	0.017 (0.78)	0.013 (1.07)
Cash	0.005 (0.50)	0.019*** (2.97)
Fla	−0.036*** (−2.79)	−0.042*** (−7.11)
Dual	0.003 (1.02)	0.003** (2.18)
Soe	−0.012*** (−2.72)	−0.002 (−0.62)
Constant	0.083 (1.33)	0.128*** (4.34)
Firm FE	YES	YES
Year FE	YES	YES
Observations	5711	15,462
R-squared	0.332	0.313

Note: T values are in parentheses. ***, ** and * denote significance at the 1 %, 5 % and 10 % levels, respectively.

4.3.3. PSM-DID

Since the LCCP policy is not a true natural experiment, there may be selection biases in the selection of pilot cities. To mitigate the potential issue of sample selection bias, this study employs propensity score matching with difference in differences (PSM-DID) for robustness testing. Ten indicators, including company size (Size), financial leverage (Lev), profitability (ROA) and the shareholding percentage of the largest shareholder (Top1), are used as covariates to construct the cross-sectional PSM. A 1:2 caliper nearest neighbor matching method is employed to calculate the matching score and to select the matched subjects. Fig. 4 displays the results of the balance test, indicating that the standardized mean differences of all matching variables are less than 5 % after matching, thus meeting the condition for balance testing. Overall, the standardized bias of the matched sample is significantly reduced, indicating a good matching effect. As shown in the first column of Table 5, using the sample that satisfies the common support assumption, the regression results of Eq. (2) remain significant at the 5 % level. This suggests that the original model does not suffer from severe sample selection bias and the conclusion that the LCCP policy reduces corporate investment efficiency is robust.

4.3.4. Alternative measure of corporate investment efficiency

To reduce the measurement error in the estimation of corporate investment efficiency, we further replace the growth opportunity indicator (Growth) in Eq. (1) with the Tobin's Q ratio. We then conduct OLS regression on Eq. (1) again and take the absolute value of the resulting residuals as the indicator for measuring corporate inefficient investment, as shown in the second column of Table 5. The regression results remain significant even after replacing the indicator for measuring corporate investment efficiency. Models constructed using different indicators yield the same conclusion: the LCCP policy significantly reduces corporate investment efficiency.

4.3.5. Excluding other policy

Apart from the LCCP policy, other policies implemented during the same period may also have an impact on the baseline regression of this study. In order to eliminate this interference, we take into account the Innovative City Pilot (ICP) policy (Zhang and Zheng, 2023) and the Carbon Emissions Trading (CET) policy (Zhang and Wang, 2021; Feng et al., 2024), and include the corresponding dummy variables in the baseline regressions. The specific regression results are shown in the third column of Table 5 and the first column of Table 6, respectively. Even when considering the impact of the ICP policy and the CET policy, the LCCP policy still significantly reduces corporate investment efficiency. The specific lists of the ICP policy and the CET policy can be found in Appendix C.

Table 8
Heterogeneity analysis II.

VARIABLES	(1)	(2)
	SOEs	Non-SOEs
	InvEff	InvEff
LCCP	−0.011*** (−3.04)	0.013** (2.26)
Size	−0.006* (−1.94)	−0.008 (−1.35)
Roa	0.116*** (4.13)	0.086** (2.13)
Lev	0.035*** (2.83)	−0.005 (−0.22)
Top1	−0.011 (−0.84)	0.018 (0.62)
Board	0.001 (0.11)	−0.003 (−0.24)
InBoard	−0.046* (−1.91)	0.036 (0.84)
Cash	0.001 (0.07)	−0.011 (−0.50)
Fla	−0.029** (−2.04)	0.003 (0.10)
Dual	−0.005 (−1.18)	0.005 (0.86)
Constant	0.200** (2.51)	0.206* (1.71)
Firm FE	YES	YES
Year FE	YES	YES
Observations	1962	1826
R-squared	0.295	0.299

Note: According to the industry classification standards issued by the China Securities Regulatory Commission in 2012, this paper classifies industries with codes C22, C25, C26, C30-C32, D44, D45, and G56 as high-carbon-emission industries. T values are in parentheses. ***, ** and * denote significance at the 1 %, 5 % and 10 % levels, respectively.

4.3.6. Sample screening

In order to mitigate the impact of outliers on the regression results, we conducted a trimming process on the samples, including trimming the top and bottom 1 % and top and bottom 5 %. The results are shown in the second and third columns of Table 6. The results indicate that the basic assumptions of this study remain valid even after excluding the influence of outliers.

5. Heterogeneity analysis

We conducted a heterogeneity analysis from the perspectives of firm diversification, ownership nature, and region, revealing the heterogeneous impact of the LCCP policy on corporate investment efficiency, thereby verifying hypothesis H2 of this study.

5.1. LCCP policy, corporate diversification and investment efficiency

The implementation of the LCCP policy has to some extent changed the external environment in which corporations operate. Corporations with a single business model do not have much room for adjustment when facing such external shocks, while corporations with a higher degree of diversification can try to avoid the negative impacts brought about by external shocks (Wang, 2023; Bai, Fu, et al., 2023). Referring to the study of Biddle et al. (2009) and Iskenderoglu (2023), we sort corporations annually based on the Herfindahl-Hirschman Index. Corporation-year observations in the bottom quartile (i.e., the smallest HH index) are classified as having a higher degree of diversification, while the other observations are classified as having a lower degree of diversification. As shown in Table 7, LCCP policy can improve the investment efficiency of corporations with a higher degree of diversification, although this effect is not significant, this conclusion is comparable to Wang (2023). At the same time, the LCCP policy significantly reduces the investment efficiency of corporations with a lower degree of diversification.

5.2. LCCP policy, ownership nature and investment efficiency

The LCCP policy requires pilot cities to adjust their economic structures, eliminate polluting processes, equipment and corporations and strictly enforce emission standards for all corporations. These requirements are bound to have a significant impact on corporations belonging to high-carbon-emission industries. To delve deeper into the influence of the LCCP policy on the investment efficiency of

Table 9
Heterogeneity analysis III.

VARIABLES	(1)	(2)
	High emissions	Low emissions
	InvEff	InvEff
LCCP	0.003** (2.17)	−0.005** (−2.00)
Size	−0.006*** (−4.05)	0.001 (0.36)
Roa	0.020 (1.53)	0.012 (0.63)
Lev	0.026*** (4.30)	0.005 (0.48)
Top1	0.020*** (2.64)	0.038*** (2.75)
Board	0.003 (0.62)	−0.008 (−1.21)
InBoard	0.022* (1.70)	−0.017 (−0.92)
Cash	0.016*** (2.59)	0.006 (0.55)
Fla	−0.042*** (−6.51)	−0.053*** (−4.86)
Dual	0.002* (1.73)	0.004* (1.76)
Soe	−0.006* (−1.72)	−0.003 (−0.67)
Constant	0.164*** (5.06)	0.063 (1.08)
Firm FE	YES	YES
Year FE	YES	YES
Observations	14,735	5462
R-squared	0.281	0.344

Note: T values are in parentheses. ***, ** and * denote significance at the 1 %, 5 % and 10 % levels, respectively.

corporations in high-carbon-emission industries with varying ownership structures. We refer to [Yu et al. \(2023\)](#) to select high-carbon-emitting industries and categorize the corporations within these industries into state-owned enterprises (SOEs) and non-state-owned enterprises (Non-SOEs) to investigate the separate impacts of the LCCP policy on their investment efficiency. As shown in [Table 8](#), within the high-carbon-emission industry, the LCCP policy significantly increases the investment efficiency of SOEs while significantly decreasing the investment efficiency of Non-SOEs, this conclusion is comparable to [Xie et al. \(2023\)](#). This suggests that the implementation of the LCCP policy significantly benefits SOEs, while Non-SOEs may suffer greater negative impacts due to information asymmetry.

5.3. LCCP policy, city carbon emissions and investment efficiency

To study the impact of the LCCP policy on cities with different carbon emission intensities, we classify corporations based on their cities' average carbon dioxide emissions in sample period. Corporation-year observations in the top quartile (i.e., Cities with the highest carbon emissions where the corporations are located) are classified as corporations located in high-carbon-emission cities, while the remaining observations are classified as corporations located in low-carbon-emission cities. As shown in [Table 9](#), the LCCP policy, as a verified policy that effectively reduces carbon dioxide emissions at the city level, significantly reduces the investment efficiency of corporations within cities with higher carbon emissions. For corporations within cities with lower carbon emissions, the LCCP policy significantly improves their investment efficiency. This result may be attributed to the difficulty for corporations located in high-carbon-emission cities to transform their carbon emission habits. As a result, the restrictions brought about by LCCP policy can have a negative impact on their investment efficiency ([Dong et al., 2023](#)). In contrast, corporations in cities with lower carbon emissions can better adapt to the policy orientation of the LCCP policy and leverage this advantage to improve their investment efficiency.

6. Further analysis

The Ministry of Ecology and Environment of the People's Republic of China released the "National Low-Carbon City Pilot Progress Assessment Report" in July 2023. The report indicates that a total of 34 cities nationwide have issued guiding documents for low-carbon development and 25 cities have issued special plans for low-carbon development. The Guangzhou Municipal People's Government issued the "14th Five-Year Plan for Low-Carbon Pilot Work" indicating a comprehensive coordination of efforts to promote

Table 10
Further analysis.

VARIABLES	(1)	(2)	(3)	(4)
	Issuing guidance	No guidance	Issuing special plans	No special plans
	InvEff	InvEff	InvEff	InvEff
LCCP	−0.004** (−1.97)	0.004*** (2.75)	−0.001 (−0.32)	0.003** (2.38)
Size	−0.006** (−2.16)	−0.004*** (−2.81)	−0.004* (−1.69)	−0.004*** (−2.81)
Roa	0.057*** (3.56)	0.008 (0.71)	0.058*** (2.81)	0.007 (0.64)
Lev	0.023** (2.55)	0.019*** (3.15)	0.022** (2.44)	0.015*** (2.77)
Top1	0.025* (1.93)	0.018*** (2.65)	0.024* (1.88)	0.022*** (3.11)
Board	−0.002 (−0.25)	0.000 (0.07)	0.014** (2.12)	−0.005 (−1.31)
InBoard	0.013 (0.62)	0.010 (0.92)	0.003 (0.18)	0.015 (1.24)
Cash	0.024** (2.23)	0.009 (1.57)	0.008 (1.07)	0.013** (2.10)
Fla	−0.045*** (−3.71)	−0.040*** (−7.14)	−0.035*** (−3.71)	−0.044*** (−7.38)
Dual	−0.000 (−0.21)	0.004** (2.55)	0.002 (0.74)	0.003** (2.06)
Soe	0.001 (0.29)	−0.007** (−2.21)	−0.002 (−0.35)	−0.005 (−1.59)
Constant	0.178*** (2.92)	0.125*** (4.40)	0.102* (1.89)	0.138*** (4.71)
Firm FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Observations	5890	16,297	6368	15,809
R-squared	0.298	0.292	0.305	0.292

Note: The cities that have issued guidance or special plans are listed in Appendix D. T values are in parentheses. ***, ** and * denote significance at the 1 %, 5 % and 10 % levels, respectively.

low-carbon pilot work and to build a batch of low-carbon pilot demonstrations within five years. In the "Long-Term Plan for Low-Carbon Development" Shenzhen explicitly proposes to achieve low-carbon development through the improvement of the low-carbon regulatory system, low-carbon industrial system and low-carbon technology support system, outlining the strategic direction for low-carbon development in Shenzhen over the next decade.

The act of local governments issuing guidance documents or special plans for low-carbon development demonstrates their strong commitment to the LCCP policy implemented by the central government. These actions help to concretize abstract policy requirements, enabling local corporations to carry out low-carbon transitions more effectively. As a result, this reduces their environmental compliance costs and decreases the environmental uncertainty they face, thereby mitigating the negative impact of the LCCP policy on corporate investment efficiency.

We distinguish the response levels of various cities to the LCCP policy based on whether each city has issued guiding documents for low-carbon development or special plans for low-carbon development. According to our analysis, the negative impact of the LCCP policy on corporate investment efficiency in cities with a high response to the LCCP policy is significantly reduced. In Table 10, columns (1) and (2) display the effects of the LCCP policy on the investment efficiency of corporations in cities with and without the issuance of guidance on low-carbon development. Columns (3) and (4) show the effects of the LCCP policy on the investment efficiency of corporations in cities with and without the issuance of special plans for low-carbon development. The regression results indicate that in the samples of columns (1) and (3), the LCCP policy has increased the investment efficiency of corporations, while in the samples of columns (2) and (4), the LCCP policy has significantly reduced the investment efficiency of corporations. This indicates that the level of the city's response to the policy significantly influences the effectiveness of LCCP policy on the investment efficiency of local corporations. Therefore, local governments can mitigate or even reverse the negative impact of this policy on the investment efficiency of local corporations by strengthening their response to LCCP policy.

7. Conclusion and policy recommendation

Our study focuses on Chinese A-share listed companies from 2009 to 2021, we find that the LCCP policy can significantly reduce city-level carbon dioxide emissions (Hou et al., 2023), but to some extent, this comes at the cost of sacrificing corporate investment efficiency. Furthermore, LCCP policy reduces corporate investment efficiency by increasing the environmental uncertainty faced by corporations, which affect normal production and operation of the firms. Heterogeneity analysis reveals that different companies in the

sample have different responses to LCCP policy, depending on diversification, ownership type and located cities. LCCP policy has a more obvious negative effect on the corporate investment efficiency of lower diversification levels, Non-SOEs within high-carbon-emission industry and those located in high-carbon emitting cities. Finally, further analysis suggests that local governments can alleviate the policy's negative impact on local corporations by increasing their response to the LCCP policy.

Based on the above conclusions, we make the following suggestions. First, policymakers should fully recognize that climate policies cannot have only positive impacts without any negative effects. Therefore, it is important to accurately and comprehensively assess the effectiveness of climate policies. Policymakers should consider providing subsidies to regions, industries, and enterprises that are more significantly impacted by the policies to help them achieve low-carbon transitions. Second, companies should fulfill their social responsibilities and actively consider how to better seize opportunities for low-carbon development in the face of the inevitable trend towards low-carbon development. Companies can expand their business scope and diversify operations to enhance their risk resistance capabilities.

Third, policy implementors should strive to concretize abstract policy requirements. Local governments can bridge the gap between theoretical requirements and practical operations by issuing guiding documents for low-carbon development or special plans for low-carbon development. This can reduce the environmental uncertainty faced by enterprises and help them achieve low-carbon transitions at minimal cost.

CRedit authorship contribution statement

Shuhan Liu: Supervision. **Zixuan Zhang:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Zhenyu Ge:** Writing – original draft, Supervision, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Low-carbon pilot city	
First Batch (2010)	Province: Hubei, Yunnan, Guangdong, Shaanxi and Liaoning City: Chongqing, Xiamen, Nanchang, Baoding, Tianjin, Shenzhen, Hangzhou and Guiyang
Second Batch (2012)	Province: Hainan City: Beijing, Shanghai, Qinhuaudao, Hulunbeier, greater Hinggan Mountains region, Huai'an, Ningbo, Nanping, Ganzhou, Jiuyuan, Guangzhou, Zunyi, Kunming, Yan'an, Shijiazhuang, Jincheng, Jilin, Suzhou, Zhenjiang, Wenzhou, Chizhou, Jingdezhen, Qingdao, Wuhan, Guilin, Guangyuan, Jinchang and Urumqi
Third Batch (2017)	City: Wuhai, Dalian, Xunke County, Changzhou, Jinhua, Hefei, Huangshan, Xuancheng, Lu'an, Gongqing, Fuzhou, Jinan, Yantai, Changsha, Chenzhou, Zhongshan, Liuzhou, Chengdu, Yuxi, Ankang, Dunhuang, Yinchuan, Wuzhong, Yining, Hotan, Alar, Shenyang, Chaoyang, Nanjing, Jiaying, Quzhou, Huaibei, Sanming, Ji'an, Weifang, Changyang Tujia Autonomous County, Zhuzhou, Xiangtan, Sanya, Qiongzong Li and Miao Autonomous County, Simao District of Pu'er, Lhasa, Lanzhou, Xining and Changji

Appendix B

Variable	Definition
Size	The natural logarithm of the company's assets at the end of the year
Lev	Ratio of total liabilities to total assets
RoA	Ratio of net profit to total assets
Fla	Ratio of current assets to total assets
Top1	Shareholding ratio of the largest shareholder
Cash	The ratio of monetary funds to total assets
Board	The natural logarithm of the number of directors
InBoard	Ratio of independent directors to number of directors
Dual	Dummy variable, if the chairman and general manager are held by the same person, it takes the value 1, otherwise the value is 0
Soe	Dummy variable, the value of state-owned enterprises is 1, otherwise the value is 0

Appendix C

Innovative pilot cities	
First batch (2008)	Shenzhen
Second batch (2010)	Beijing, Tianjin, Shijiazhuang, Tangshan, Shenyang, Dalian, Shanghai, Nanjing, Changzhou, Wuxi, Suzhou, Ningbo, Jiaxing, Hangzhou, Fuzhou, Xiamen, Jinan, Qingdao, Yantai, Guangzhou, Haikou, Taiyuan, Changchun, Harbin, Hefei, Baotou, Nanning, Chongqing, Nanchang, Jingdezhen, Zhengzhou, Luoyang, Chengdu, Guiyang, Kunming, Xi'an, Wuhan, Changsha, Baoji, Lanzhou, Yinchuan, Xining, Changji
Third batch (2011)	Qinhuangdao, Lianyungang, Zhenjiang, Hohhot
Fourth batch (2012)	Nantong, Urumqi
Fifth batch (2013)	Taizhou, Yangzhou, Yancheng, Huzhou, Jinan, Pingxiang, Nanyang, Xiangyang, Yichang, Zunyi
sixth batch (2018)	Xuzhou, Shaoxing, Jinhua, Quanzhou, Longyan, Weifang, Dongying, Foshan, Dongguan, Ma'anshan, Wuhu, Zhuzhou, Hengyang, Jilin, Yuxi, Lhasa
Carbon emission trading pilot area	
First batch (2011)	Beijing, Tianjin, Shanghai, Shenzhen, Chongqing, Guangdong, Hubei
Second batch (2013)	Fujian

Appendix D

Cities with high responsiveness to LCCP policy	
issued special guidance	Qinhuangdao, Hulunbuir, Zhenjiang, Nanping, Kunming, Urumqi, Shenyang, Chaoyang, Changzhou, Zhuzhou, Chenzhou, Zhongshan, Yuxi, Xining, Changji, Nanchang, Guiyang, Jincheng, Suzhou, Wuhan, Guangyuan, Yan'an, Wuhai, Lu'an, Ankang, Qiongzong, Sanming, Liuzhou, Nanjing, Xiangtan, Hangzhou, Shenzhen, Qingdao, Ningbo, Guilin, Dunhuang, Yinchuan, Pu'er, Hefei, Changsha
Issued guiding documents	Nanchang, Kunming, Tianjin, Jiyan, Yan'an, Ningbo, Zunyi, Jinchang, Baoding, Qinhuangdao, Urumqi, Qiongzong Li and Miao Autonomous County, Chaoyang, Hefei, Huaibei, Xuancheng, Changsha, Ankang, Dunhuang, Gongqingcheng, Jinan, Zhuzhou, Guilin, Jinhua, Weifang, Guangyuan, Guangzhou, Shijiazhuang, Qingdao, Ji'an, Yantai, Yinchuan, Beijing, Quzhou

Data availability

Data will be made available on request.

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